

1. Administrative & Study Identification

Full Study Title: An Investigational Immunotherapy Study of BMS-986258 Alone and in Combination With Nivolumab in Participants With Solid Cancers That Are Advanced or Have Spread
Official Title: A Phase 1/2 First-in-Human Study of BMS-986258 Alone and in Combination With Nivolumab in Advanced Malignant Tumors
Short Title/Acronym: CA031-002 **Protocol Number:** CA031-002
NCT Number: NCT03446040 **Sponsor/Company Name:** Bristol Myers Squibb
Investigational Product: BMS-986258 (Anti-TIM-3 Monoclonal Antibody), Nivolumab (Anti-PD-1; Trade Names: Opdualag, Opdivo; ATC Code: L01FF01), and Recombinant human hyaluronidase PH20 (rHuPH20)
Phase of Development: PHASE1, PHASE2 **Medical Monitor:** Bristol Myers Squibb Clinical Trials (clinical.trials@bms.com)

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To determine whether BMS-986258 both as a monotherapy and in combination with Nivolumab is safe and tolerable in the treatment of advanced malignant tumors.
Secondary Objectives: To evaluate the Objective Response Rate (ORR), Median Duration of Response (mDOR), and Progression-Free Survival (PFS) Rate at 6, 9, and 12 Months.

2.2 Background & Rationale
The immune checkpoint receptor TIM-3 (T cell immunoglobulin and mucin domain-containing protein 3) is commonly overexpressed in multiple types of cancer and is associated with T-cell exhaustion and tumor immune evasion. Inhibiting TIM-3 with BMS-986258 aims to rejuvenate the anti-tumor response of CD8+ cytotoxic T-lymphocytes. Given that single-agent PD-1 axis interventions may eventually face acquired resistance, the combination of anti-TIM-3 (BMS-986258) with an established anti-PD-1 agent (Nivolumab) aims to tackle immunorefractory tumors by concurrently targeting alternative immune checkpoint pathways to improve clinical outcomes.

3. Study Design & Methodology

3.1 Design Framework
Study Type: Interventional **Control Type:** Single-arm / Dose Escalation & Cohort Expansion
Blinding: Open-label (None) **Randomization:** Non-Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration
Targeted Enrollment: 92 participants
Number of Sites: Multicenter (e.g., US, Canada, Australia, Japan)
Estimated Study Start: September 2018
Trial Status: TERMINATED (Study was terminated early)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologic confirmation of one of 5 tumors: Renal Cell Carcinoma (RCC), Colorectal Cancer (CRC), Non-Small Cell Lung Cancer (NSCLC), Squamous Cell Carcinoma of the Head and Neck (SCCHN), or Triple Negative Breast Cancer (TNBC) that is metastatic, recurrent, and/or unresectable. 2. Measurable disease per Response Evaluation Criteria in Solid Tumors v1.1 (RECIST v1.1). 3. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 4. Participants must have received, and then progressed, relapsed, or been intolerant to, at least 1 standard treatment regimen in the advanced or metastatic setting according to solid tumor histologies. 5. Women must agree to follow specific methods of contraception, if applicable.

4.2 Exclusion Criteria 1. Active, known, or suspected autoimmune disease. 2. Treatment with cytotoxic agents, unless at least 4 weeks have elapsed from the last dose of prior anti-cancer therapy and the initiation of study therapy. 3. Other active malignancies requiring concurrent intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered via Dose Escalation and Cohort Expansion arms. * **Part A (Dose Escalation):** BMS-986258 monotherapy. * **Part A1:** BMS-986258 + Recombinant human hyaluronidase PH20 (rHuPH20). * **Part B (Dose Escalation):** BMS-986258 + Nivolumab. * **Part C (Cohort Expansion):** BMS-986258 + Nivolumab. **Route of Administration:** Intravenous (IV) and Subcutaneous (SC - for the rHuPH20 cohort). **Duration of Treatment:** Until disease progression (per RECIST v1.1) or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care as clinically indicated. **Prohibited:** Concurrent anti-cancer/cytotoxic agents, active systemic steroid therapies, and any investigational agents not specified by the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy verification, Baseline Imaging
Baseline	Day 0	Treatment assignment, First Dose Administration, Baseline Labs
Interim	Standard Treatment Cycles	Investigational product administration, Safety & PK/ADA monitoring, Tumor Imaging (e.g., Q8W)
End of Study	Follow-up	Final Vital Signs, Toxicity resolution check, Long-term Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. **Number of Participants With Adverse Events (AEs):** Any new untoward medical occurrence or worsening of a preexisting condition. 2. **Number of Participants Who Died During the Study.** 3. **Number of Participants With Dose Limiting Toxicities (DLTs):** Side effects severe enough to prevent an increase in dose level. **Measurement Method:** Clinical assessment using CTCAE criteria.

7.2 Secondary Endpoints * **Objective Response Rate (ORR):** Evaluated per RECIST v1.1. * **Median Duration of Response (mDOR)** * **Progression-Free Survival (PFS) Rate at 6, 9, and 12 Months**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of any unfavorable and unintended signs (such as an abnormal laboratory finding), symptoms, or diseases. **Serious Adverse Events (SAEs):** Reported immediately if they result in death, are life-threatening, require inpatient hospitalization, or result in persistent or significant disability/incapacity. **Data Monitoring Committee (DMC):** Safety and toxicity data evaluated continuously by sponsor/internal safety review teams to guide dose-escalation limits.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All treated subjects are included in the Safety Set. Efficacy assessments are performed on participants with measurable disease at baseline. **Sample Size Justification:** A planned targeted enrollment of 92 participants driven by dose-escalation rules followed by statistical powering for the specified cohort expansion tumor types. **Handling of Missing Data:** Efficacy markers like PFS are tracked up to the date of first objectively documented disease progression or death, relying on Kaplan-Meier estimates to handle censored data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical site monitoring to ensure protocol adherence, accurate reporting of DLTs, and proper RECIST v1.1 application. **Audit Trail:** Robust tracking of eCRF, laboratory and imaging data via independent review, and safety board checkpoints.

11. Study Identifiers & Governance

Primary ID: CA031-002 **Registry Number:** NCT03446040 **Posting ID:** 354035027163764280 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** An Investigational Immunotherapy Study of BMS-986258 Alone and in Combination With Nivolumab in Participants With Solid Cancers That Are Advanced or Have Spread **Trial Status:** TERMINATED (Study was terminated early)

12. Clinical Framework (Oncology Specific)

Disease Areas / Primary Condition: Advanced Cancer (Advanced, Metastatic, or Unresectable Solid Tumors: RCC, CRC, NSCLC, SCCHN, TNBC) **Clinical Classifications & Terminology:** * **UMLS Preferred Name:** Immunotherapy of Neoplasms * **MeSH Code:** D009369 * **SNOMED CT:** 416629007, 406858000, 254637007 **Diagnosis Threshold:** Histologic or cytologic confirmation of malignancy with measurable disease. **Medical Methodology:** Immune Checkpoint Inhibition (ICI) combination therapy. **Optimization Target:** Exhausted CD8+ cytotoxic T-lymphocyte rejuvenation to overcome innate or acquired immunotherapy resistance.

13. Study Procedures & Methodology

Management Pathway: Sequential dose-escalation determining the Maximum Tolerated Dose (MTD)/Recommended Phase 2 Dose (RP2D), followed by cohort expansion. **Education Protocol (HCP):** Site initiation visits covering AE/SAE management, immune-related adverse event (irAE) handling, and RECIST evaluation. **Education Protocol (Patient):** Informed Consent and Study Participant's Guide detailing potential immunotherapy side effects. **Quality Audit Mechanism:** Centralized evaluation of safety thresholds, radiographic tumor assessments, and PK/ADA laboratory tracking.

14. Observation & Follow-Up Schedule

Total Duration: Treatment until disease progression, plus ongoing survival follow-up phases. **Onsite Visit Cadence:** Dictated by treatment cycles (typically Q2W, Q3W, or Q4W for infusions). **Remote Monitoring Frequency:** Regular safety follow-ups via phone or clinic if off-treatment but awaiting progression/survival milestones. **Checklist Review Interval:** Regular radiographic tumor imaging required to satisfy RECIST v1.1 parameters.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |